Sources of variability in the human word response

Research proposal (draft)

Juee Naik

November 5, 2026

Gap

Large iEEG studies of free recall describe the average response to a word: high gamma rises first in visual cortex and later in prefrontal cortex (Burke et al., 2014; Kucewicz et al., 2019), and theta, alpha/beta and gamma power are induced at largely separate sites, in separate sequences ("mosaic"; 139 subjects; Marks et al., 2021). These studies average over trials and treat electrodes as samples of a region; Marks et al. (2021) also normalize each trial within its own epoch rather than against a pre-word baseline. How much of the response differs between subjects, electrodes, sessions and trials, whether its spectral components are independent beyond space, and what changes it from trial to trial have not been measured. With 362 subjects, 1009 sessions and ∼300 words per session, this dataset can answer these questions. Memory effects are out of scope (see DeHaan, 2026).

Aims

Aim 1. Where does the variability sit? Partition the word response (word on vs. the blank before it) into subject, electrode, session and trial components per region, over time and frequency, and estimate how reliable each level is. *Preliminary (40 subjects; Fig. 1):* single trials are dominated by trial-to-trial variation (71–94% outside occipital cortex, 43% in occipital high gamma), of which task structure (serial position, list number, blank length, same category as the previous word) explains only 3–7% of the variance once its chance fit is removed. In the trial-averaged response, electrodes carry most of the variance in occipital cortex (∼60–70%) and the subject share is largest in limbic regions and the hippocampus (20–30%).

Aim 2. What is the stable part? Describe the electrode-level structure: response types from each electrode's spectral and temporal profile, their anatomy and latency, and whether a type is a fixed property of an electrode across sessions. *Preliminary:* five types emerge (Fig. 2), from broadband increases (34% of occipital electrodes) to alpha/beta decreases with late high gamma and low-up/high-down profiles (frontal, parietal); an electrode's pattern replicates between session halves (Fig. 4; median $r = 0.90$ occipital high gamma).

Aim 3. Are the spectral components independent beyond space, and what moves them from trial to trial? Marks et al. (2021) found the bands separate across sites. *Preliminary:* they are also independent trial by trial at the same electrode (Fig. 3; median r between -0.04 and 0.06 in every lobe, unchanged with task structure removed). Serial position scales the high-gamma response without shifting it (Figs. 5, 6), alpha habituates across lists (Fig. 8), and the blank interval has small occipital effects (Fig. 9) that can be tested against recovery from adaptation (Groen et al.,

2022) vs. temporal expectation. Planned: single-trial latency, baseline vs. gain, and the same tests per response type.

Open issues

- Trial-level artifact rejection must be added to the main pipeline (one artifact trial distorts Fig. 10); the preliminary analyses drop robust outlier trials.
- Per-trial time–frequency power is not cached (only alpha, high gamma and trial-averaged spectra), so the full time × frequency × level partition needs a recompute.
- The session share includes the sampling noise of each session mean; split-half reliability will separate it. Type names are read off one clustering run and are provisional.

Figures (most to least relevant)

Status in brackets: *prelim.* = 40 random subjects; *full* = all subjects; *paper* = from the word-onset paper.

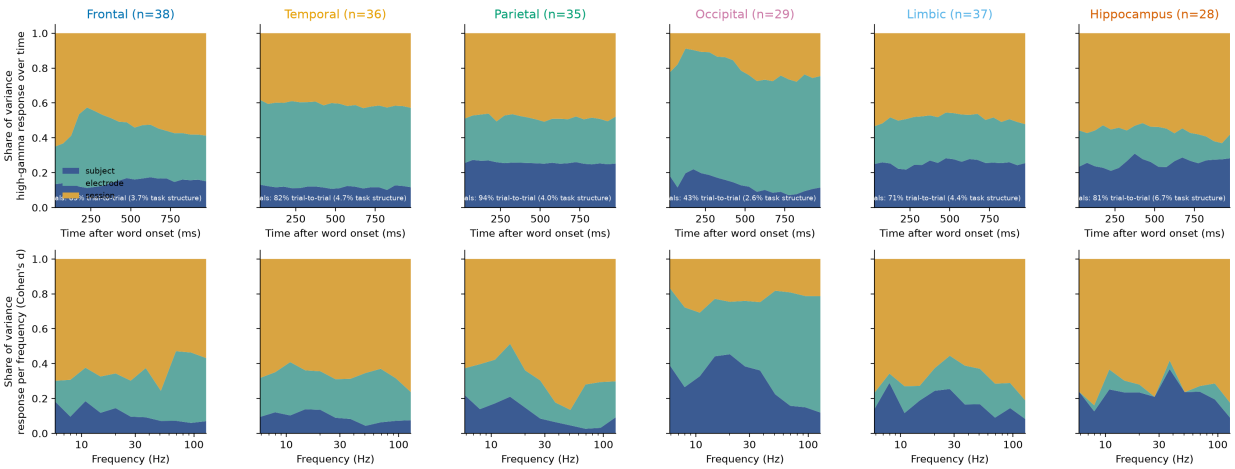


Figure 1: [prelim.] Variance partition of the word response per lobe. Top: high-gamma response over time (trial-averaged per electrode-session; the trial-to-trial and task-structure shares of single trials in white). Bottom: response per frequency (Cohen's d, 11 bins 5–150 Hz). Subject (blue), electrode within subject (teal), session within electrode (yellow, including sampling noise).

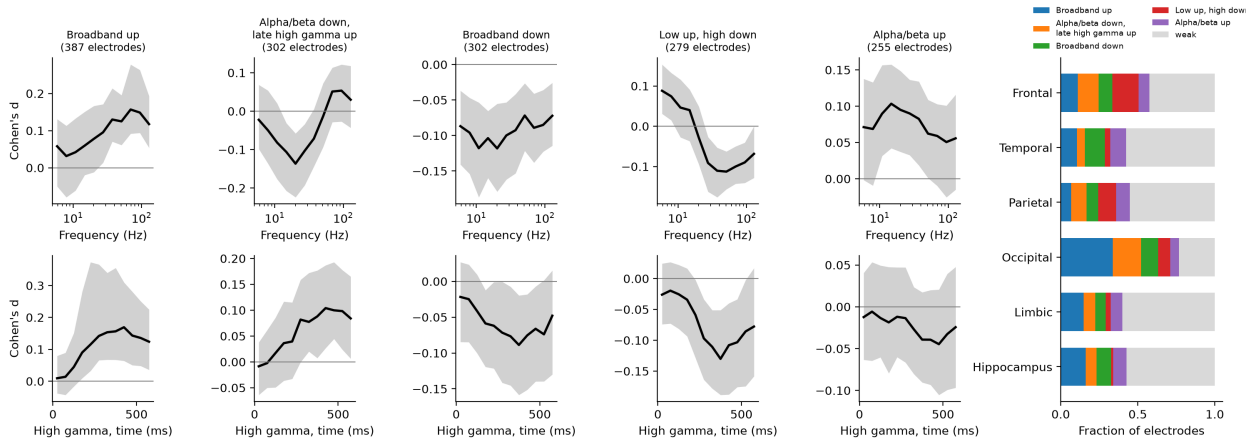


Figure 2: [prelim.] Electrode response types (k-means on spectral and high-gamma time profiles of the more responsive half of electrodes): median (line) and interquartile range per type, and each lobe's composition.

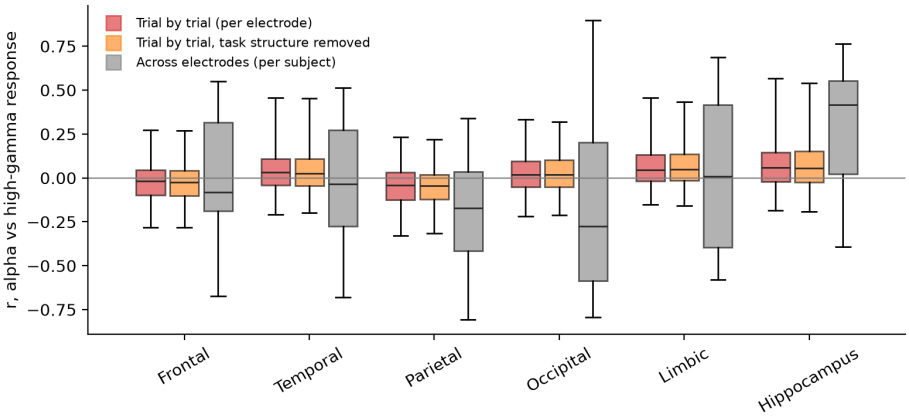


Figure 3: [prelim.] Alpha vs. high-gamma response: Pearson r across trials at each electrode (red), the same with task structure regressed out (orange), and across electrodes within each subject (grey).

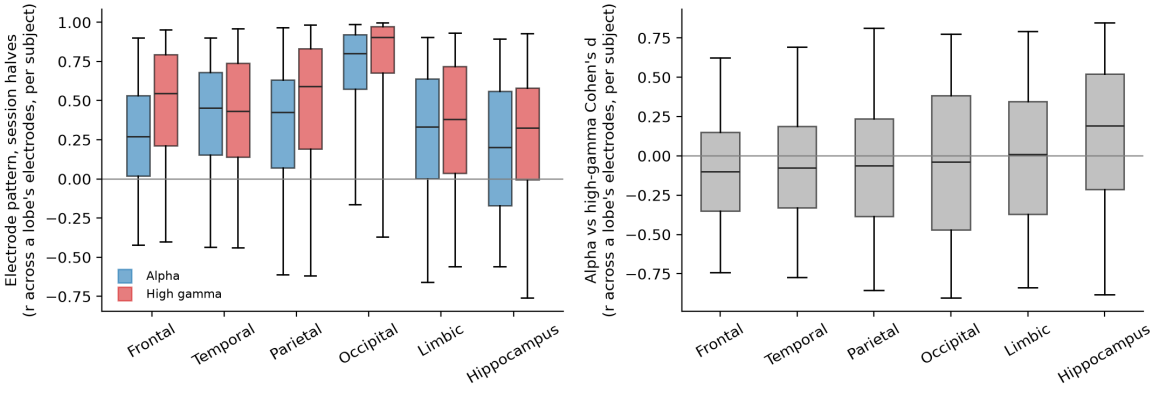


Figure 4: [full] Left: an electrode's response pattern replicates between session halves (per subject and lobe, r across electrodes). Right: alpha vs. high-gamma Cohen's d across electrodes.

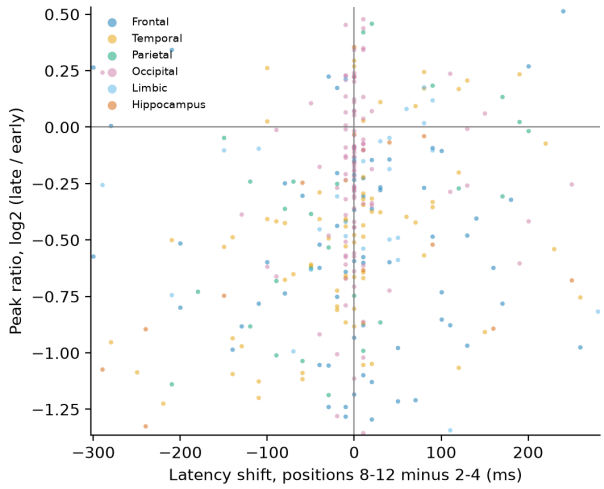


Figure 5: [prelim.] Gain vs. timing: per responsive electrode, high-gamma peak ratio (late / early in the list) against the latency shift.

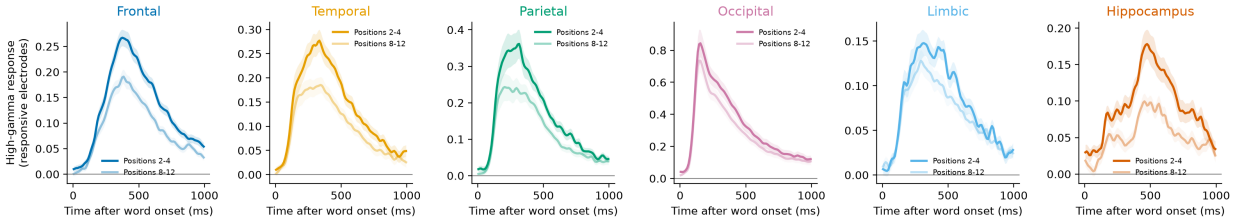


Figure 6: [full] High-gamma response for serial positions 2–4 (dark) and 8–12 (light): smaller, not later.

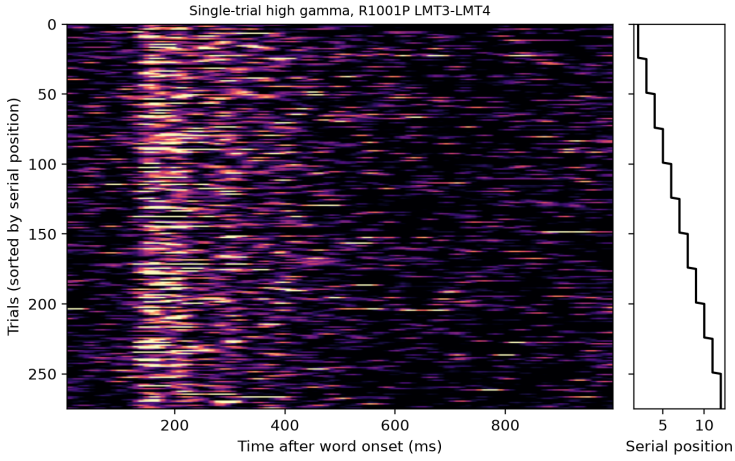


Figure 7: [prelim.] Single-trial high gamma of one occipital electrode, trials sorted by serial position.

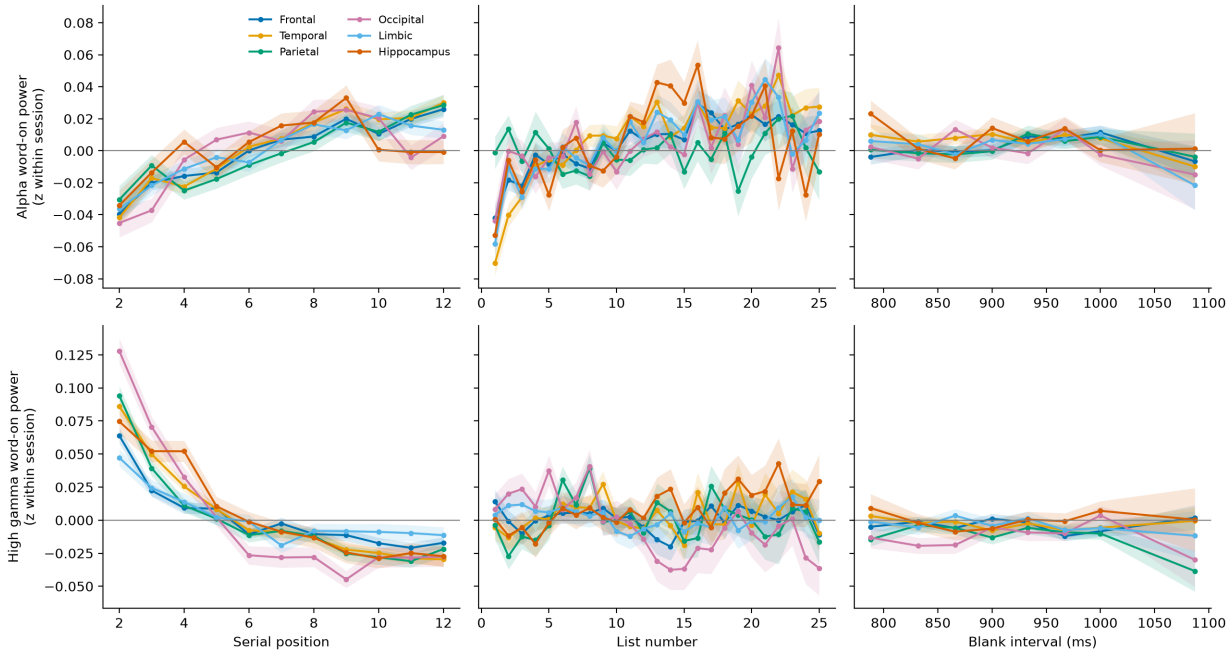


Figure 8: [full] Word-on power by serial position, list number and blank interval per lobe.

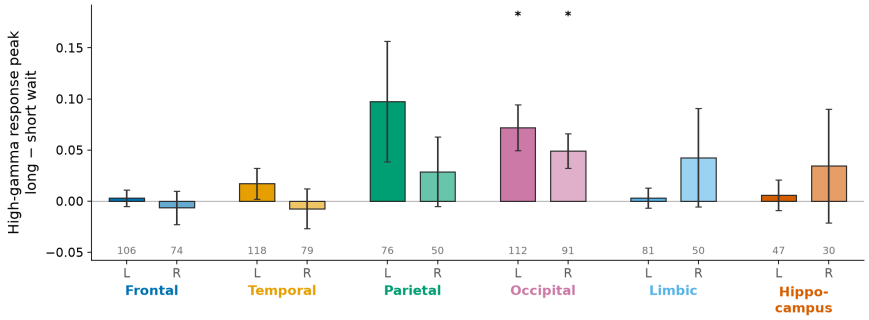


Figure 9: [paper] Occipital high gamma after long vs. short blank intervals.

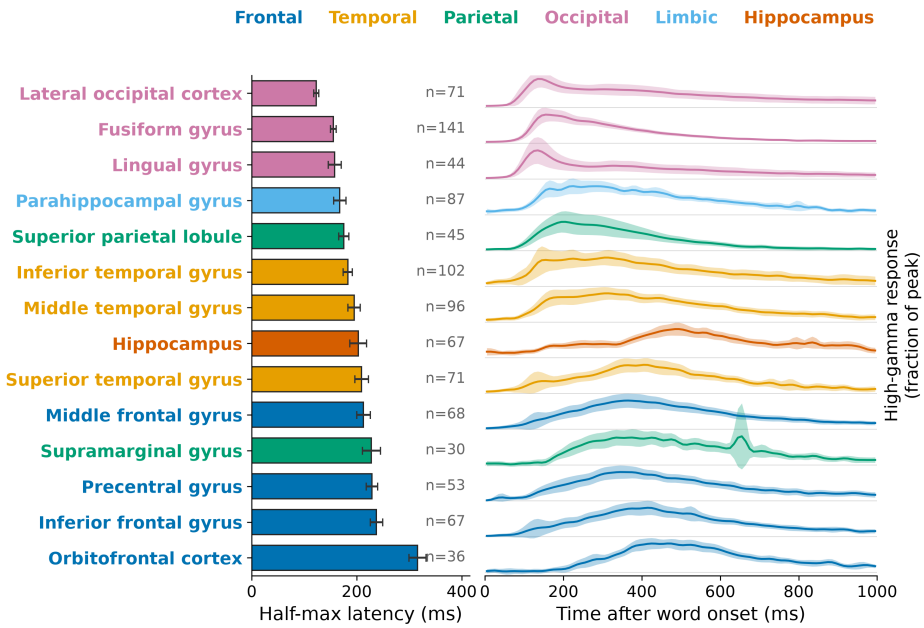


Figure 10: [paper] High-gamma latency per region (the supramarginal spike at 650 ms is one artifact trial).

References

Burke, J. F., Long, N. M., Zaghloul, K. A., Sharan, A. D., Sperling, M. R., and Kahana, M. J. (2014). Human intracranial high-frequency activity maps episodic memory formation in space and time. *NeuroImage*, 85:834–843.

DeHaan, R. D. (2026). *On the specificity, causality, and reliability of electrophysiological correlates of human memory: limitations and opportunities*. PhD thesis, University of Pennsylvania.

Groen, I. I. A., Piantoni, G., Montenegro, S., Flinker, A., Devore, S., Devinsky, O., Doyle, W., Dugan, P., Friedman, D., Ramsey, N. F., Petridou, N., and Winawer, J. (2022). Temporal dynamics of neural responses in human visual cortex. *The Journal of Neuroscience*, 42(40):7562–7580.

Kucewicz, M. T., Saboo, K., Berry, B. M., Kremen, V., Miller, L. R., Khadjevand, F., Inman, C. S., Wanda, P., Sperling, M. R., Gorniak, R., Davis, K. A., Jobst, B. C., Lega, B., Sheth, S. A., Rizzuto, D. S., Iyer, R. K., Kahana, M. J., and Worrell, G. A. (2019). Human verbal memory encoding is hierarchically distributed in a continuous processing stream. *eneuro*, 6(1):ENEURO.0214–18.2018.

Marks, V. S., Saboo, K. V., Topçu, Ç., Lech, M., Thayib, T. P., Nejedly, P., Kremen, V., Worrell, G. A., and Kucewicz, M. T. (2021). Independent dynamics of low, intermediate, and high frequency spectral intracranial eeg activities during human memory formation. *NeuroImage*, 245:118637.